



Module Wise Question Bank

Module 2: Data Link Layer and Medium Access Control Sub layer

<i>Qn No.</i>	<i>Question</i>	<i>RBTL</i>	<i>CO</i>	<i>Marks</i>
1	Explain the design issues of the Data Link Layer	L2	CO2	10
2	Define framing and mention any two framing methods.	L2	CO2	10
2	Demonstrate the difference between byte stuffing and bit stuffing along with an example	L2	CO2	6
3	Explain the flow control in data link layer. List the various approaches of flow control.	L2	CO2	10
4	Explain the need of error control in data link layer. List and explain types of errors along with appropriate diagram	L2	CO2	10
5	Explain the purpose of error detection in the Data Link Layer	L2	CO2	6
6	Explain the parity check error detection technique with a diagram. Compare the odd parity and even parity.	L2	CO2	10
7	Explain the CRC technique used for error detection with a diagram. Mention the advantage of CRC over parity check	L2	CO2	10
8	Compare parity, CRC, and Hamming Code in terms of error detection and correction capabilities.	L2	CO2	6
9	Explain Stop-and-Wait ARQ with a timing diagram. What are its limitations?	L2	CO2	10
10	Explain Go-Back-N ARQ protocol with sender and receiver window diagrams.	L2	CO2	10
11	Discuss Selective Repeat ARQ protocol. How does it improve efficiency over Go-Back-N?	L2	CO2	10
12	Compare Stop-and-Wait, Go-Back-N, and Selective Repeat ARQ in terms of efficiency, complexity, and buffer requirements.	L2	CO2	10
13	Explain Pure ALOHA and Slotted ALOHA. Derive throughput efficiency for each.	L2	CO2	10
14	Explain CSMA, CSMA/CD, and CSMA/CA protocols. Highlight their working principles and differences.	L2	CO2	10
15	With a diagram explain the IEEE 802.3(Ethernet) frame format	L2	CO2	10
16	Explain the working of Ethernet using CSMA/CD. Mention advantages and limitations.	L2	CO2	10
17	Discuss the architecture of IEEE 802.11 Wireless LAN. Differentiate between Infrastructure and Ad-hoc modes.	L2	CO2	10
18	Draw and explain the IEEE 802.11 MAC frame format.	L2	CO2	10
19	Explain how collisions are avoided in IEEE 802.11 networks.	L2	CO2	10
20	Problem on Parity, CRC and Hamming Code	L2	CO2	10